THE DEBRIEF · FREE EDITION

Your code. Better understood.

C++ practice · 11/1/2026

Automatic multiple-choice results and written-answer self-assessments are reported separately. These are informal study records, not official grades. Your code was parsed locally, never compiled or executed.

Automatically graded

— No MC answers

0 correct / 0 recorded multiple-choice questions

Exact answer-key comparison. Skipped MC questions count as incorrect. Written answers never contribute to this score.

Written self-assessment

4	1	0	0
Understood	Partially	Not yet	Unassessed

Chosen by you after reviewing the reference. This is not an objective correctness grade.

Concepts practiced

5 answered / 5 main questions · 0 skipped

modulo · if · variable · initialization · return

Saved question-order seed: interface-review

Your next study session

Review **modulo**. Explain the cited statement, check its assumptions, and compare with the reference checklist.

Weak topics come from incorrect/skipped MC answers or your written/follow-up self-assessments.

Question by question

QUESTION 1 · modulo

Self-assessed: partial

Is using a zero divisor with built-in % valid?

Source lines 9–9

```
        int digit = temp % 10;
```

Your response

Using a zero divisor is not valid for built-in integer remainder. I would check that it is nonzero first.

Reference explanation

No. Remainder with a zero divisor has undefined behavior; validate the divisor before evaluation.

Expected points to check

No. Remainder with a zero divisor has undefined behavior; validate the divisor before evaluation.

Relate this rule to the cited source and state any relevant assumptions.

QUESTION 2 · if

Self-assessed: complete

When does the consequence of the if on line 5 execute?

Source lines 5–5

```
    if (!(cin >> number) || number < 100 || number > 999) return 1;
```

Your response

I would explain the language rule and relate it to the cited source.

Reference explanation

When its condition, contextually converted to bool, is true.

Expected points to check

When its condition, contextually converted to bool, is true.

Relate this rule to the cited source and state any relevant assumptions.

QUESTION 3 · variable

Self-assessed: complete

Does changing a copied primitive variable automatically change the original? Refer to number.

Source lines 4–4

```
    int number;
```

Your response

I would explain the language rule and relate it to the cited source.

Reference explanation

Ordinary primitive value copies are separate objects; modifying one does not modify the other. Verify whether the declaration is actually a value or a reference.

Expected points to check

- Ordinary primitive value copies are separate objects; modifying one does not modify the other. Verify whether the declaration is actually a value or a reference.
- Relate this rule to the cited source and state any relevant assumptions.

QUESTION 4 · initialization

Self-assessed: complete

For a primitive local value initialized from another value, when is that initial value captured?

Source lines 6–6

```
    int temp = number;
```

Your response

I would explain the language rule and relate it to the cited source.

Reference explanation

The initializer is evaluated when execution reaches the declaration; later changes to a copied source value do not automatically update the copy.

Expected points to check

- The initializer is evaluated when execution reaches the declaration; later changes to a copied source value do not automatically update the copy.
- Relate this rule to the cited source and state any relevant assumptions.

QUESTION 5 · return

Self-assessed: complete

Can a non-void function generally reach its closing brace without returning a value safely?

Source lines 15–15

```
    return 0;
```

Your response

I would explain the language rule and relate it to the cited source.

Reference explanation

Flowing off the end of a value-returning function generally has undefined behavior; main has a special rule equivalent to returning zero at its end. Ensure each relevant path returns appropriately.

Expected points to check

- Flowing off the end of a value-returning function generally has undefined behavior; main has a special rule equivalent to returning zero at its end. Ensure each relevant path returns appropriately.
- Relate this rule to the cited source and state any relevant assumptions.